

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 October 2026
Team ID	SWTID-2026-3491
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2

Example: Employee data import using Import Sets and Transform Maps in ServiceNow

Solution Architecture - Spreadsheet Import using Transform Maps (ServiceNow)

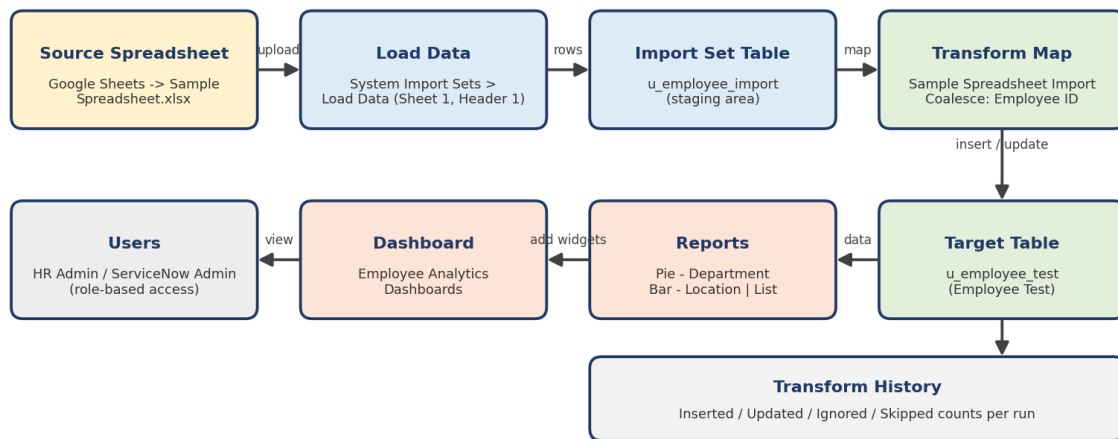


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application	ServiceNow platform UI (Application Navigator, forms, lists)
2.	Application Logic-1	Loading spreadsheet rows into the staging table	Load Data / Import Set (u_employee_import)
3.	Application Logic-2	Mapping and transforming staged rows into the target table	Transform Map "Sample Spreadsheet Import" with Auto Map and Mapping Assist
4.	Application Logic-3	Duplicate prevention on repeat imports	Coalesce on Employee ID field map
5.	Database	Employee Import (staging) and Employee Test (target) tables with String fields	ServiceNow relational database
6.	Cloud Database	Database service on cloud	ServiceNow cloud instance

S.No	Component	Description	Technology
7.	File Storage	Source spreadsheet attached at import	Sample Spreadsheet.xlsx uploaded through Load Data
8.	External API-1	Not used in this project	N/A
9.	External API-2	Not used in this project	N/A
10.	Machine Learning Model	Not used in this project	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on cloud	ServiceNow cloud instance (no local server required)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Spreadsheet prepared in Google Sheets / Excel; platform itself is ServiceNow	Google Sheets, Microsoft Excel (.xlsx)
2.	Security Implementations	Role-based access; ACL on pa_dashboards (Create) with Admin Overrides; reports shared with chosen users, groups and roles	ServiceNow ACLs, roles, Share options
3.	Scalable Architecture	Staging-table pattern separates loading from transformation, so large spreadsheets can be processed in bulk	Import Sets + Transform Maps
4.	Availability	Hosted on the ServiceNow cloud instance, accessible from any browser	ServiceNow cloud
5.	Performance	Coalesce ignores unchanged rows (4 rows ignored on identical re-import); Transform History shows inserted / updated / ignored counts	Transform Map, Transform History

References:

- <https://c4model.com/>
- <https://www.ibm.com/cloud/architecture>
- <https://aws.amazon.com/architecture>